

Assignment 1

Find out the 10 providers and write down their last 12 months released model along with data- model name, released date, cut off date, input cost, output cost, context window, model type, use of it.

Provider Sr.No.	Provider	Model Sr.No.	Model	Release Date	Knowledge Cutoff	Cost (Input)	Cost (Output)	Context Window	Model Type	When to Use
1	OpenAI	1	GPT Chat Latest	May-2026	Oct-2023	\$5	\$30	400K	Public	Default production chat workloads; auto-upgraded latest model without migration effort
		5	GPT-5.5 Pro	Apr-2026	Dec-2025	\$30	\$180	1.05M	Public	Mission-critical reasoning, complex workflows, agentic execution
		6	GPT-5.5	Apr-2026	Dec-2025	\$5	\$30	1.05M	Public	Advanced enterprise workloads with balanced cost vs performance
		7	GPT-5.4 Image 2	Apr-2026	Aug-2025	\$8	\$15	272K	Public	Multimodal workflows (text + image generation)
		8	GPT-4o	Apr-	Aug-2025	\$0.60 chars		4K	Public	Low-cost text-to-speech

			Mini TTS	2026		-				systems
		10	GPT-5.4 Nano	Mar-2026	Aug-2025	\$0.20	\$1.25	400K	Public	High-throughput, low-latency pipelines (classification, routing)
		11	GPT-5.4 Mini	Mar-2026	Aug-2025	\$0.75	\$4.5	400K	Public	Scalable production systems requiring efficiency
		12	GPT-5.4 Pro	Mar-2026	Aug-2025	\$30	\$180	1.05M	Public	Deep reasoning, long-context enterprise automation
		13	GPT-5.4	Mar-2026	Aug-2025	\$2.5	\$15	1.05M	Public	General-purpose flagship for coding + reasoning
		14	GPT-5.3 Chat	Mar-2026	Aug-2025	\$1.75	\$14	128K	Public	Conversational AI, assistants, chat platforms
		15	GPT-5.3 Codex	Feb-2026	Aug-2025	\$1.75	\$14	400K	Public	End-to-end software development workflows
		16	GPT Audio	Jan-2026	Oct-2023	\$2.5	\$10	128K	Public	Audio-based applications with natural voice output
		17	GPT Audio Mini	Jan-2026	Oct-2023	\$0.60	\$2.4	128K	Public	Cost-sensitive voice applications
		18	GPT-5.2 Codex	Jan-2026	Aug-2025	\$1.75	\$14	400K	Public	Coding agents, debugging, large-scale code automation

	19	GPT-5.2	Dec-2025	Aug-2025	\$1.75	\$14	400K	Public	General-purpose AI with adaptive reasoning
	20	GPT-5.2 Pro	Dec-2025	Aug-2025	\$21	\$168	400K	Public	High-accuracy, high-stakes problem solving
	21	GPT-5.1	Nov-2025	Sep-2024	\$1.25	\$10	400K	Public	Balanced general AI workloads
	22	GPT-5.1 Codex	Nov-2025	Sep-2024	\$1.25	\$10	400K	Public	Developer-centric coding workflows
	23	GPT-5	Aug-2025	Sep-2024	\$1.25	\$10	400K	Public	Foundational enterprise AI adoption
	24	GPT-5 Mini	Aug-2025	May-2024	\$0.25	\$2	400K	Public	Lightweight applications, fast response systems
	25	GPT-5 Nano	Aug-2025	May-2024	\$0.05	\$0.40	400K	Public	Ultra-low latency, real-time systems
	26	o3 Pro	Jun-2025	Jun-2024	\$20	\$80	200K	Public	Advanced reasoning-heavy workloads
	27	o4 Mini	Apr-2025	Jun-2024	\$1.10	\$4.40	200K	Public	Cost-efficient reasoning + tool usage
	28	o3	Apr-2025	Jun-2024	\$2	\$8	200K	Public	Strong reasoning across STEM and coding
	29	GPT-4.1	Apr-2025	Jun-2024	\$2	\$8	1.05M	Public	Long-context reasoning, enterprise knowledge systems

		30	GPT-4.1 Mini	Apr-2025	Jun-2024	\$0.40	\$1.60	1.05M	Public	Cost-optimized general AI applications
		31	GPT-4.1 Nano	Apr-2025	Jun-2024	\$0.10	\$0.40	1.05M	Public	High-speed, low-cost micro tasks

2	Anthropic	1	Claude Sonnet 4	May-2025	Mar-2025	\$3 / M	\$15	1M	Public	General enterprise AI, coding assistants, Knowledge work, Everyday Professional tasks
		2	Claude Opus 4.1	Aug-2025	Mar-2025	\$15 / M	\$75 / M	200K	Public	Precisionheavy coding, Multifile refactoring, Toolassisted reasoning, Data analysis
		3	Claude Sonnet 4.5	Sep-2025	Jul-2025	\$3	\$15	1M	Public	AI agents, Workflow orchestration, Financial analysis, Longcontext automation
		4	Claude Haiku 4.5	Oct-2025	Feb-2025	\$1 / M	\$5	200K	Public	Fast chatbots, Realtime assistants, Highvolume inference, Costsensitive workloads
		5	Claude Opus 4.5	Nov-2025	Aug-2025	\$5 / M	\$25	200K	Public	Frontier reasoning, Autonomous research agents, Browser/computeruse AI, Multistep planning

		6	Claude Opus 4.6	Feb-2026	May-2025	\$5 / M	\$25	1M	Public	Enterprisegrade coding, Large codebase understanding, Technical architecture planning, Longsession coherence
		7	Claude Sonnet 4.6	Feb-2026	Aug-2025	\$3	\$15	1M	Public	Balanced enterprise AI, Computeruse workflows, Project management copilots, Document generation
		8	Claude Opus 4.6 Fast	Apr-2026	May-2025	\$30	\$150	1M	Public	Ultrafast premium inference, Timesensitive enterprise workflows, Largescale asynchronous agents
		9	Claude Opus 4.7	Apr-2026	Jan-2026	\$5	\$25	1M	Public	Persistent autonomous agents, Endtoend project orchestration, Multistage debugging, Strategic reasoning
		10	Claude Opus Latest	Apr-2026	Jan-2026	Varies	Varies	1M	Public	Always use latest Opus generation, Best for adaptive production systems
		11	Claude Sonnet Latest	Apr-2026	Aug-2025	Varies	Varies	1M	Public	Production applications needing stable latest Sonnet model

		12	Claude Haiku Latest	Apr-2026	Feb-2025	Varies	Varies	200K	Public	Lightweight production apps, Fast customerfacing assistants
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3	Perplexity	1	Sonar	Jan-2025	Near real-time web	\$1	\$1	127K	Public	Fast AI search, lightweight Q&A, customer support assistants
		2	Sonar Pro	Mar-2025	Near real-time web	\$3	\$15	200K	Public	Enterprise search, multi-step research, long-form retrieval
		3	Sonar Reasoning Pro	Mar-2025	Near real-time web	\$2	\$8	128K	Public	Advanced reasoning, analytical workflows, citation-heavy responses
		4	Sonar Deep Research	Mar-2025	Near real-time web	\$2	\$8	128K	Public	Autonomous research, market analysis, comprehensive report generation
		5	Sonar Pro Search	Oct-2025	Near real-time web	\$3	\$15	200K	Public	Agentic search systems, AI analysts, multi-step knowledge workflows
		6	Embed V1 0.6B	Mar-2026	N/A	\$0.004 usage pricing	N/A	32K	Private	Lightweight embeddings, low-latency vector search, semantic retrieval
		7	Embed V1 4B	Mar-2026	N/A	\$0.03 usage pricing	N/A	32K	Private	High-quality embeddings, enterprise RAG, large-scale semantic search

4	Meta	1	Llama Guard 3 8B	Feb-2025	Dec-2023	\$0.48	\$0.03	131K	Private	AI safety moderation, prompt/response filtering
		2	Llama 4 Maverick	Apr-2025	Aug-2024	\$0.15	\$0.60	1.05M	Private	Advanced multimodal AI, long-context reasoning, image understanding
		3	Llama 4 Scout	Apr-2025	Aug-2024	\$0.08	\$0.30	328K	Private	Efficient multimodal assistants, multilingual visual AI
		4	Llama Guard 4 12B	Apr-2025	Aug-2024	\$0.18	\$0.18	164K	Private	Multimodal content moderation, AI safety systems

5	xAI	1	Grok 3 Beta	Apr-2025	Feb-2025	\$3	\$15	131K	Public	Enterprise AI, structured reasoning, coding, summarization
		2	Grok 3 Mini Beta	Apr-2025	Feb-2025	\$0.30	\$0.50	131K	Public	Lightweight reasoning, math tasks, fast logical inference
		3	Grok 3	Jun-2025	Feb-2025	\$3	\$15	131K	Public	Data extraction, coding, finance/legal analysis, enterprise assistants
		4	Grok 3 Mini	Jun-2025	Feb-2025	\$0.30	\$0.50	131K	Public	Cost-efficient reasoning, quantitative analysis, lightweight agents

		5	Grok 4	Jul-2025	Jul-2025	\$3	\$15	256K	Public	Advanced reasoning, multimodal AI, structured outputs, parallel tool calling
		6	Grok Code Fast 1	Aug-2025	Sep-2025	\$0.20	\$1.50	256K	Public	Agentic coding, debugging, developer copilots
		7	Grok 4 Fast	Sep-2025	Sep-2025	\$0.20	\$0.50	2M	Public	Large-context AI, cost-efficient enterprise assistants, fast multimodal workflows
		8	Grok 4.1 Fast	Nov-2025	Sep-2025	\$0.20	\$0.50	2M	Public	Customer support AI, deep research, scalable AI agents
		9	Grok 4.20	Mar-2026	Sep-2025	\$1.25	\$2.50	2M	Public	High-accuracy reasoning, enterprise workflows, agentic tool usage
		10	Grok 4.20 Multi-Agent	Mar-2026	Sep-2025	\$2	\$6	2M	Public	Multi-agent collaboration, deep research orchestration, autonomous workflows
		11	Grok 4.3	Apr-2026	Dec-2025	\$1.25	\$2.50	1M	Public	Long-document analysis, deep research, advanced agentic reasoning

6	DeepSeek	1	DeepSeek V4 Pro	Apr-2026	Apr-2026	\$0.435	\$0.87	1.04M	Public	advanced reasoning, coding, and long-horizon agent workflows, with strong performance across knowledge, math, and software engineering benchmarks.
		2	DeepSeek V3.2	Dec-2025	Aug-2025	\$0.57	\$1.71	131K	Hybrid	Reasoning-heavy AI systems, mathematical problem solving
		3	DeepSeek V4 Flash	Apr-2026	Apr-2026	\$0.14	\$0.28	1.04M	Hybrid	fast inference and high-throughput workloads, while maintaining strong reasoning and coding performance.
7	NVIDIA	1	Nemotron 3 Nano Omni	Apr-2026	Nov-2025	Free	Free	256K	Yes (Open)	Best for multimodal enterprise agents handling text, audio, video, and images together. Useful for AI copilots, video analysis, smart surveillance, and multimodal RAG systems.

		2	Nemotron 3 Super	Mar-2026	Jun-2025	Free	Free	1M	Yes (Open)	Ideal for long-context reasoning, autonomous agents, coding assistants, enterprise workflows, and multi-document analysis. Strong fit for advanced AI orchestration.
		3	Llama Nemotron Embed VL 1B V2	Feb-2026	Jun-2025	Free	Free	131K	Yes (Open)	Use for multimodal embeddings, semantic search, RAG pipelines, document retrieval, OCR search, and visual knowledge bases.
		4	Nemotron 3 Nano 30B A3B	Dec-2025	Nov-2025	Free	Free	256K	Yes (Open)	Best for lightweight enterprise AI agents, private deployments, efficient reasoning, and cost-sensitive production AI systems.
		5	Nemotron Nano 12B 2 VL	Oct-2025	May-2025	Free	Free	128K	Yes (Open)	Excellent for OCR, charts, document intelligence, image reasoning, and video understanding workloads. Useful in banking, healthcare, and analytics automation.

		6	Nemotron Nano 9B V2	Sep-2025	May-2025	Free	Free	32K	Yes (Open)	Good for lightweight reasoning, summarization, chatbots, math reasoning, and local/private AI deployments with lower hardware requirements.
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8	Alibaba /Qwen	1	Qwen3.5 Plus 2026-04-20	Apr-2026	Jan-2026	\$0.40	\$2.40	1M	Public	Long-context multimodal reasoning, enterprise AI agents, video + image analysis
		2	Qwen3.6 Flash	Apr-2026	Jan-2026	\$0.25	\$1.50	1M	Public	Fast inference, chatbot workloads, scalable AI assistants
		3	Qwen3.6 Max Preview	Apr-2026	Feb-2026	\$1.04	\$6.24	262K	Public	Agentic coding, deep reasoning, autonomous workflows
		4	Qwen3.6 27B	Apr-2026	Jan-2026	\$0.60	\$3.60	262K	Private	Coding copilots, repository understanding, multilingual reasoning
		5	Qwen3.6 Plus	Apr-2026	Jan-2026	\$0.325	\$1.95	1M	Public	SWE-bench coding, 3D/game generation, advanced reasoning
		6	Qwen3.5-35B-A3B	Feb-2026	Dec-2025	\$0.1625	\$1.30	256K	Hybrid	Cost-efficient multimodal tasks, reasoning + vision

		7	Qwen3.5-27B	Feb-2026	Dec-2025	\$0.195	\$1.56	256K	Hybrid	Balanced inference speed and multimodal reasoning
		8	Qwen3.5-122B-A10B	Feb-2026	Dec-2025	\$0.26	\$2.08	262K	Hybrid	Advanced text + image understanding, enterprise reasoning
		9	Qwen3.5-Flash	Feb-2026	Dec-2025	\$0.065	\$0.26	1M	Public	Budget AI workloads, high-throughput APIs
		10	Qwen3.5 Plus 2026-02-15	Feb-2026	Nov-2025	\$0.26	\$1.56	1M	Hybrid	Enterprise assistants, RAG pipelines
		11	Qwen3.5 397B A17B	Feb-2026	Nov-2025	\$0.39	\$2.34	256K	Hybrid	SOTA multimodal reasoning, GUI interaction, agentic AI
		12	Qwen3 Max Thinking	Feb-2026	Nov-2025	\$0.78	\$3.90	262K	Public	Complex reasoning, scientific workflows, advanced agents
		14	Qwen3 VL 32B Instruct	Oct-2025	Jul-2025	\$0.104	\$0.416	262K	Public	OCR, document AI, video understanding
		15	Qwen3 VL 8B Thinking	Oct-2025	Jul-2025	\$0.117	\$1.365	256K	Private	Visual reasoning, STEM image analysis
		16	Qwen3 VL 8B Instruct	Oct-2025	Jul-2025	\$0.117	\$0.455	256K	Private	VQA, OCR, GUI control

		17	Qwen3 VL 30B A3B Thinking	Oct-2025	Jun-2025	\$0.13	\$1.56	262K	Hybrid	Advanced multimodal reasoning, visual coding
		18	Qwen3 VL 30B A3B Instruct	Oct-2025	Jun-2025	\$0.13	\$0.52	262K	Hybrid	Production multimodal apps, OCR systems
		19	Qwen3 VL 235B A22B Thinking	Sep-2025	Jun-2025	\$0.26	\$2.60	131K	Hybrid	Research-grade multimodal reasoning
		20	Qwen3 VL 235B A22B Instruct	Sep-2025	Jun-2025	\$0.26	\$1.04	131K	Hybrid	Enterprise document AI, multimodal assistants
		21	Qwen3 Max	Sep-2025	Jun-2025	\$0.78	\$3.90	256K	Private	Enterprise reasoning, multilingual AI
		22	Qwen3 Coder Plus	Sep-2025	Jun-2025	\$0.65	\$3.25	128K	Public	Autonomous coding agents, software engineering
		23	Qwen3 Coder Flash	Sep-2025	Jun-2025	\$0.195	\$0.975	128K	Public	Fast code generation, CI/CD integrations
		24	Qwen3 Next 80B A3B	Sep-2025	May-2025	\$0.0975	\$0.78	262K	Hybrid	Long-chain reasoning, math proofs

			Thinking							
		25	Qwen3 Next 80B A3B Instruct	Sep-2025	May-2025	\$0.0975	\$0.78	262K	Hybrid	RAG, tool-use workflows, enterprise chatbots
		26	Qwen Plus 0728	Sep-2025	May-2025	\$0.26	\$0.78	1M	Hybrid	Long-context enterprise assistants
		27	Qwen3 30B A3B Thinking 2507	Aug-2025	Apr-2025	\$0.13	\$1.56	131K	Public	Structured reasoning, competitive coding
		28	Qwen3 Coder 30B A3B Instruct	Jul-2025	Apr-2025	\$0.2925	\$1.463	256K	Public	Repository-scale coding, agentic tool use
		29	Qwen3 30B A3B Instruct 2507	Jul-2025	Apr-2025	\$0.13	\$0.52	131K	Public	Instruction following, multilingual AI
		30	Qwen3 235B A22B Thinking 2507	Jul-2025	Apr-2025	\$0.1495	\$1.495	262K	Public	Advanced reasoning, research, AI agents
		31	Qwen3 Coder	Jul-2025	Apr-2025	\$0.975	\$4.875	1.05M	Hybrid	Massive-scale code generation, enterprise

			480B A35B							engineering
		32	Qwen3 235B A22B Instruct 2507	Jul-2025	Apr-2025	\$0.1495	\$0.598	262K	Hybrid	Multilingual reasoning, math + coding
		33	Qwen3 30B A3B	Apr-2025	Dec-2024	\$0.13	\$0.52	131K	Private	General reasoning, agent tasks
		34	Qwen3 8B	Apr-2025	Dec-2024	\$0.117	\$0.455	131K	Private	Lightweight assistants, multilingual chat
		35	Qwen3 14B	Apr-2025	Dec-2024	\$0.2275	\$0.91	132K	Private	Coding + reasoning balanced workloads
		36	Qwen3 32B	Apr-2025	Dec-2024	\$0.104	\$0.416	131K	Private	Enterprise assistants, multilingual reasoning
		37	Qwen3 235B A22B	Apr-2025	Dec-2024	\$0.455	\$1.82	131K	Hybrid	Frontier open-source reasoning
9	Amazon Bedrock	1	Nova Pro 1.0	Dec-2025	Oct-2024	\$0.80	\$3.20	300K	Public	Visual question answering (TextVQA) and video understanding (VATEX), Financial documentation

		2	Nova Micro 1.0	Dec-2025	Oct-2024	\$0.035	\$0.14	128K	Public	Text summarization, translation, content classification, interactive chat, and brainstorming. mathematical reasoning and coding abilities.
		3	Nova Lite 1.0	Dec-2025	Oct-2024	\$0.06	\$0.24	300K	Public	Real-time customer interactions, document analysis, and visual question-answering tasks with high accuracy
		4	Nova Premier 1.0	Oct-2925	NA	\$2.50	\$12.50	1M	Public	Complex reasoning,best teacher for distilling custom models
		5	Nova 2 Lite	Dec-2025	NA	\$0.30	\$2.50	1M	Public	Generate text from video, image,Processing documents, Generating code

10	Google	1	Gemini 3.1 Pro Preview	Feb-2026	Jan-25	\$2	\$12	1.04M	Public	Advanced reasoning, agentic workflows, SWE tasks, multimodal enterprise AI
		2	Gemini 3 Pro Preview	Nov-2025	Shutdown	\$2	\$12	1.04M	Public	Frontier reasoning, coding agents, research synthesis, multimodal planning

		3	Gemini 2.5 Pro	Jun-2025	Jan-25	\$1.25	\$10	1.04M	Public	Complex reasoning, mathematics, coding, scientific tasks
		4	Gemini 2.5 Flash Lite	Jul-2025	Jan-25	\$0.10	\$0.40	1.04M	Public	High-throughput inference, low latency chatbots, cheap automation
		5	Gemini 2.0 Flash	Jan-2025	Jan-25	\$0.10	\$0.40	1.04M	Public	Fast multimodal assistants, function calling, agents
		6	Gemini 2.0 Flash Lite	Feb-2025	Jan-25	\$0.075	\$0.30	1.04M	Public	Budget inference, large-scale automation
		7	Gemma 4 31B	Apr-2026	NA	\$0.13	\$0.38	262K	Private	Open-source reasoning, multilingual AI, coding
		8	Gemma 4 26B A4B	Apr-2026	NA	\$0.08	\$0.35	262K	Private	Efficient MoE inference, multimodal applications
		9	Gemma 3 27B	Dec-2025	NA	\$0.08	\$0.16	131K	Private	Open multimodal assistants, reasoning, coding
		10	Gemma 3 12B	Dec-2025	NA	\$0.04	\$0.13	131K	Private	Mid-size local AI deployments
		11	Gemma 3 4B	Dec-2025	NA	\$0.04	\$0.08	128K	Private	Lightweight local inference, edge AI
		12	Nano Banana 2	Feb-2026	Jan-25	\$0.50	\$3	65k	Public	AI image generation and editing
		13	Nano Banana	Nov-2025	Jan-25	\$2	\$12	66K	Public	Professional-grade image synthesis and editing

			Pro							
		14	Lyria 3 clip preview	Mar-2026	NA	\$0.04 per song		1.04M	Public	Music generation model
		15	Lyria 3 pro preview	Mar-2026	NA	\$ 0.08 per song		1.04M	Public	Music generation model
		16	Veo 3.1	Mar-2026	N/A	Video-based	From \$0.40/sec	N/A	Public	Cinematic AI video generation
		17	Veo 3.1 Fast	Apr-2026	N/A	Video-based	From \$0.10/sec	N/A	Public	Faster AI video production
		18	Veo 3.1 Lite	Apr-2026	N/A	Video-based	From \$0.05/sec	N/A	Public	Bulk short-video generation
		19	Chirp 3	May-2026	N/A	Audio	\$0.016/m in	N/A	Public	Speech transcription, call-center AI, meeting notes

Assignment 2:

Transformer Article:

- Article 1 <https://jalammar.github.io/illustrated-transformer/>
- Article 2 <https://jalammar.github.io/illustrated-bert>
- Article 3 <https://huggingface.co/papers/>
- Article 4 <https://nlp.seas.harvard.edu/annotated-transformer>
- Article 5 <https://sebastianraschka.com/blog/2023/llm-reading-list.html>
- Article 6 <https://jalammar.github.io/how-gpt3-works-visualizations-animations>